

Armand Le Douarec

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Education

University of Oxford, UK — MSc Quantum Technologies	2026 – 2027
Imperial College London, UK — Exchange Year, BSc Theoretical Physics & Mathematics	2025 – 2026
81 ECTS across BSc Mathematics/Physics, MSc Quantum Dynamics/Quantum Fields & Fundamental Forces	
Key modules: Quantum Information, Quantum Field Theory, General Relativity, Graph Theory, Dynamical Systems	
Average grade : 75.7 % (First-class)	
EPFL - École Polytechnique Fédérale de Lausanne, Switzerland — BSc Physics	2023 – 2026
Grade 5.55 / 6.00 (top 3–5 %)	
National Academic recognition : Swiss Study Foundation Fellow	
Lycée Saint-Louis de Gonzague (Franklin), Paris — Maths & Physics - Grade: 16.62/20, highest honours	2020 – 2023

Research

Research & Development Intern, Relational Intelligence — <i>Categorical quantum AI for NLP</i>	06/2026 – 10/2026
- Justified the categorical-AI architecture for photonic hardware, using the Geometry of Interaction (GoI) to represent recursive language processing as optical feedback. - Proved convergence bounds and maximal expressivity for stationary photonic streams. - Developed a certified method to differentiate stationary photonic streams using implicit differentiation and phase-shift measurements, avoiding layer-by-layer backpropagation through the loop.	
Research Student, Imperial College London — <i>Research presented at QALYPSO 2026</i>	🐙 2025–present
- Built an end-to-end hybrid quantum–classical AMR prototype for 2D MHD, combining a fourth-order solver, an interpretable Ising Hamiltonian, QAOA and Bayesian training across eight instability tests. - Found topology-aware gains within individual scenarios, then used leave-one-scenario-out falsification to show that the physical feature set, not QAOA depth, limits generalisation.	

Experience

Software & Product Engineer, Giedi-Labs (<i>stealth UAV startup</i>)	07/2025 – 10/2025
- Developed an AI surrogate model to reject aerodynamically non-viable UAV configurations before CFD, focusing compute on high-potential designs. - Optimised the pipeline by coupling parametric CAD (CadQuery) to automated meshing and OpenFOAM only for final validation, halving design-cycle time. - Led a three-person development team for the frontend, REST API, database/production deployment.	
Teaching Assistant, Advanced Analysis / Calculus II, EPFL	02/2025 – 07/2025
- Led weekly sessions in differential equations and multivariable calculus for first-year physics students - Developed guided problem-solving explanations and adapted them to different levels.	
Computational Modelling in C++	🐙 2023 – 2025
- Built RK4 and semi-explicit solvers with visuals for gas dynamics, the 3-body problem & satellite orbits. - Quantified chaos (using Lyapunov-like indicators) examining system behavior for accuracy & robustness.	
IBM Quantum Developer Track — 100+ hours: Qiskit, QML (kernel, classifier), VQE and hardware execution	2025

Technical Skills

Quantum	Photonic · Boson Sampling · Qiskit · QAOA / VQE / QSP · QML · block encodings · surface codes
Category theory	Optyx · ZX-calculus · DisCoCat and higher-order DisCoCat · Geometry of Interaction
ML / HPC	PyTorch · NumPy / SciPy · Optuna / CVXPY · MHD / AMR · surrogate modelling · OpenFOAM
Software	Python (advanced) · C++ (numerical) · JavaScript · TypeScript · MATLAB · Git · REST APIs · $\LaTeX$
Languages	French (native) · English (C1, IELTS : 8/9) · German (basic)

Interests & Volunteering

Coach, EPFL (09/2024 – 07/2025) — mentored 10+ first-year physics students.

Sports: basketball, karate (1st dan), sailing, skiing, hiking, running.